

CHAPTER 2

REVIEW OF LITERATURE

This chapter reviews forty distinct studies grouped into five subtopics: evaluation protocol and anomaly taxonomies (4), flow features and benchmark datasets (5), reconstruction-based anomaly detection (4), graph and relational detection (14), and drift, adversarial evasion and operational evaluation (13, including user and entity behavior analytics and explanation). Counts are by primary subtopic; studies cross-cited where literatures overlap keep a single home. The nine team-supplied papers resolve as follows: the EULER review into [17], the structural-attack review into [25] (same work as the Venturi paper), the graph neural network systematic review into [24], XAI-IDR into [40] (the Excel's P12 and P33 entries are that one Panchal and colleagues paper, single count), the pre-training review into [20], the Local Interpretable Model-agnostic Explanations and Shapley Additive Explanations review into [36], the user-behaviour-analytics review into [39] (same work as the published review, single count), the human-centred review into [38], and the R+R review into [23] (camera-ready title "R+R: From Claims to Crashes", cited via arXiv:2503.20281). Packet-generation, neuro-symbolic, ATT&CK-automation, failure-aware and tech-report adaptation studies beyond the host-graph field, and the temporal-graph pair (a measured negative for edge-level alerts), are out of scope and not counted. Also out of scope and not counted: the general anomaly and outlier surveys (background only), the kernel-boundary baseline, the certified-smoothing study (supervised classifiers, no transfer), the two preprint studies and the low-weight tail-risk study, the future-pillar datasets (no project measurement taken yet), the SIEM and Doc2Vec user and entity behavior analytics studies (kept user and entity behavior analytics representation is the review [39]), the MMD-coreset study (kept continual representation is SSF plus ReCDA), the sketch-based provenance study, the Internet-of-Things-hardening study, the preprint evasion detector, synthetic-trace generation, the insider-hybrid study, hybrid deep-classical learners, the evasion taxonomies and reviews, and the prior-draft team papers on tree explanations, syscall sequences and host self, which the nine-paper set above supersedes. GraphIDS and ARGUS are held provisional and Mvula is excluded as mistaken identity. Local citations below run [1]-[40] in primary-home order; cross-cited studies keep a single home. The mapping table at the end gives the full references.

Reading note: figures attributed to bracketed citations are surveyed results reported by those studies; unattributed figures are present-work measurements quoted as contrast against the surveyed literature, not as surveyed results themselves. Figures that could not be pinned to a source page are withheld rather than quoted approximately. The protocol group fixes the vocabulary the rest of the chapter uses. The dataset group establishes which releases can support relational modelling at all. The reconstruction group centres on Isolation Forest, the SNU variational report, the MLSDA autoencoder and the Kitsune ensemble, with PCA as the

linear reference. The graph group centres on GraphSAGE, graph convolutional networks and host-graph detectors built on them, extended by the 28-study systematic review, few-shot pre-training, the temporal-walk system, three provenance advanced persistent threat detectors, and a structural-attack-versus-hardening exchange. The final group centres on drift adaptation, continual learning, timing and structural evasion, certification, explanation and user and entity behavior analytics studies, which supply the evaluation controls the present work adopts. Each section below keeps one subtopic together for several long paragraphs so that papers addressing the same problem are compared on problems first, then on methodologies, then on systems and reported performances, rather than being scattered across short subsections.

2.1 Theoretical Background of Intrusion Detection and Anomaly Taxonomies

The organising problem of this literature is that a detector built to recognise known patterns has no principled path to an alert on behaviour it has never seen, and the early taxonomies state this consequence in plain terms. The taxonomy in [1] separates misuse detection, which matches stored signatures, from anomaly detection, which models normality and flags deviation; the companion discussion in [2]-[3] records that signature systems decide by rule match, so a family absent from the rule set can only be caught by accident. Both agree on the resulting performance picture: misuse systems report high precision on covered families and near-zero recall on anything new, while anomaly systems trade precision for coverage and shift the hard problem to threshold setting and false-alarm control. The asymmetry is structural rather than incidental: a signature either matches or it does not, so its errors concentrate in false negatives on novelty, while a normality model scores everything on a continuum, so its errors concentrate in false positives on benign rarity. Any evaluation that reports accuracy on imbalanced captures without separating these error kinds credits the misuse system for the background majority while hiding the anomaly system's queue burden; the held-out protocol with rank-based reporting exists to deny both illusions at once. A third class, specification-based detection, checks behaviour against hand-written rules and inherits the labour cost of manual specification without the learning ability of either sibling; it appears here only to complete the taxonomy, since no hand-written rule set could cover the encrypted, high-throughput traffic of the evaluation captures. The earliest proof that the anomaly side of this split works came from host sequences rather than networks: normal patterns of operating-system behaviour were learned and deviations flagged as intrusions, establishing the learn-normality premise every study in this chapter inherits, from flow autoencoders to host-graph detectors. The cross-domain survey in [3] formalises point, contextual and collective anomalies and shows that network intrusions are collective almost by definition, since a single flow from a scanner is individually ordinary and only the fan-out across a window marks the behaviour; it is the premise for the host-window scoring unit adopted here, one machine during one time slice rather than one flow in isolation.

Read together, [2], [1] and [3] set the constraint this project accepts, namely that the detector must be unsupervised, benign-only, and judged on families it never saw. The cost of the constraint is accepted explicitly: anomaly systems absorb the false-positive burden that misuse systems avoid, so every design decision after this section, from percentile calibration to the dual-gated response policy, exists to pay that burden down without abandoning the coverage that justifies it. A project that chose the misuse side would report cleaner numbers on covered families and have nothing to say about novelty; this one reports noisier numbers on unseen families and nothing else, which is the honest side of the split for zero-day work.

The strongest objection to laboratory numbers came from the closed-world critique, which argued that results measured inside one capture collapse once the stationarity assumption behind them breaks. The study in [4] reports that machine-learning network intrusion detection system work routinely trains and tests on the same capture with the same background mix, tunes thresholds where they are measured, and lets the metric absorb class imbalance instead of fixing the protocol, so published accuracy describes memorisation of one background rather than detection of new behaviour. Its methodology is a review of evaluation practice rather than a new detector, and its system-level recommendation is blunt: test on behaviour the model never saw, state the alert unit before quoting any metric, and treat threshold selection as part of the method under test. The performances it re-examines fall once these controls are applied, which is the finding this project reproduced directly when training on attack-day benign traffic instead of Monday-only benign traffic dropped mean receiver operating characteristic area under the curve from 0.9300 to 0.8405, confirming that an attack day's benign half is not clean normality. The response adopted here follows [4] together with the evaluation warnings in [3]: each of the seven CICIDS2017 families is held out in turn, the detector trains only on benign traffic, thresholds are fixed on validation, and every number is a four-seed graphics-processing-unit-deterministic band, which is stricter than random train-test splits and is the closest honest proxy for a zero-day available in this data.

Threshold setting deserves its own paragraph because it is where most surveyed numbers silently inflate. A threshold tuned where it is measured borrows information from the test distribution: the operating point is chosen after seeing which attacks arrive, so the reported accuracy describes a detector that already knows the exam questions. The honest alternative fixes the threshold on benign validation plus a held-out attack slice, then freezes it before the test families are ever scored. That discipline costs headline points and buys transferability, which is the trade this project accepts throughout. Window sizing interacts with the same mechanism through the collective-anomaly framing of [3]: a sixty-second window maximises precision at one hundred because attacker neighbourhoods stand out against small benign populations, while eighteen-hundred-second windows raise mean area under the curve by smoothing benign behaviour at the price of diluting the attacker into a larger crowd. Neither window is universally correct; the choice is a metric trade stated explicitly, with sixty seconds as the default because the alert queue, not the curve, is the operational object. Compared on problems, [4] diag-

noses protocol leakage while [3] diagnoses unit mismatch; compared on methodologies, frozen thresholds plus fixed windows are the two controls that stop laboratory numbers from describing memorisation; compared on reported performances, every number in Chapter 5 carries both controls, which is why the headline bands are narrower than single-seed single-window claims in the surveyed literature.

2.2 Theoretical Background of Flow Features and Benchmark Datasets

The benchmark papers address a shared problem: captures differ in age, realism, labelling density and, critically, in whether they can support relational modelling at all, so dataset choice preselects the detector class before any training begins. The CICIDS2017 generation paper in [5] describes a five-day capture with benign background plus seven attack families and roughly 2.8 million flows (International Conference on Information Systems Security and Privacy 2018, pages 108-116); its system ships as two releases under one name that are not interchangeable, a 79-column release carrying no Internet Protocol columns that can only feed per-flow models and an 85-column release carrying Flow ID, source and destination Internet Protocol addresses, transport protocol and Timestamp, encoded latin-1 rather than UTF-8, that alone can build host graphs. The Thursday WebAttacks file holds roughly 459,000 rows of which only about 170,000 are labelled, the remainder carrying Not-a-Number addresses that break any host-set construction unless dropped explicitly: a single Not-a-Number host makes ordered set construction raise a type error mid-sweep, which silently removes a whole family and leaves a mean across seven families computed over six with nothing marking the absence. The files are latin-1 encoded rather than untransformed text, so naive readers either crash or mangle the non-ASCII bytes in address and timestamp fields; the project reader handles the encoding explicitly. The follow-up capture in [6] repeats the warning in a new form: nine of its ten processed files again lack Internet Protocol columns and only one file can build a graph, while a third column-naming convention silently zeroes byte-count features that assume either of the first two. The usable file carries the dataset's own misspelling in its official name, a small warning that string matching against expected filenames fails on real releases. The naming drift is not cosmetic: column vocabularies differ ("Total Length of Forward Packets" against "Total Length of Forward Packets" against abbreviated machine-generated headers, with one official filename itself misspelt), so any feature pipeline that matches columns by string rather than by content silently computes zeros. The schema-mapping utility built for this project identifies columns by their value distributions instead, which is the content-based answer to a vocabulary problem the literature documents but rarely solves. The botnet-capture study in [7] contributes thirteen scenarios with background traffic plus labelled Neris, Rbot and Virut families (doi 10.1016/j.cose.2014.05.011), and the UNSW-NB15 paper in [8] contributes a hybrid synthetic-plus-real capture with nine family labels (Military Communications and Information Systems 2015, Canberra, doi 10.1109/MilCIS.2015.7348942). The training discipline for [7] is back-

ground flows strictly before the first botnet-labelled flow, so no attack-adjacent traffic leaks into the normality model, and the documented synthetic generation of [8], rather than hidden background composition, is what makes transfer results on it interpretable: when numbers move between captures, the difference can be attributed to documented generation choices instead of unknown background shifts. In reported use, [5] underpins nearly all held-out-family studies including this one, and [7] with [8] supply the botnet and hybrid counterpoints; a paper trained only on the 79-column release of [5] cannot have tested a host graph whatever its title suggests, which is the comparison filter applied throughout this chapter.

The extractor and feature literature addresses the companion problem that raw flow counters mislead twice over, once through port and address handling and once through scaling. The CICFlowMeter documentation in [9] defines more than eighty per-flow statistics covering packet-length summaries, inter-arrival-time summaries, flag counts and segment sizes, and its system output became the de facto vocabulary in which the detectors of [5] and [8] are expressed; early port-based classification work reviewed alongside [9] showed ports alone failing once ephemeral randomisation appeared, which is why the present work treats destination port as one modelled feature among seventy-six rather than as a reliable key. The four output groups have different statistical characters and different blindnesses: packet-length summaries separate bulk transfer from interactive traffic but cannot see timing; inter-arrival-time summaries carry rate and burstiness but cannot see volume; flag counts mark protocol-state abuse but fire rarely; segment-size features reflect path properties but vary with infrastructure rather than intent. Any detector leaning on one group inherits that group's blindness, and the surveyed evasions each target exactly one group: timing reshaping crosses timing-weighted detectors, tunnelling normalises flags, bulk-hiding defeats length-only rules. The seventy-six-feature input used here exists so that crossing the detector requires defeating all four groups at once, which is the feature-level analogue of the structural price list in Section 2.4. Count features add the scaling failure: byte and packet totals follow a power-law distribution, so plain min-max scaling maps the busiest host to 1.0 and compresses every other host toward zero, letting a few servers permanently own the top of any ranked queue. The measured correction in this project is logarithmic compression before scaling, which lifted mean precision at 100 from 0.250 to 0.413 with Patator moving from 0.000 to 0.618 and WebAttacks from 0.000 to 0.381, and that single preprocessing choice outweighs any architectural widening tried.

The generation methodology behind [5] explains why these pathologies cluster in one capture family. Benign traffic is synthesised from behavioural profiles while attacks are scripted across the five days, so the background is realistic in volume but narrow in habit: the same users, the same services, the same diurnal rhythm. Any detector that memorises background habit rather than attack structure inherits this narrowness, which is exactly what the attack-day contamination experiment demonstrates. The health-check procedure this project copies from the credible [5] users is therefore mechanical: report usable rows, dropped rows, graph counts and host counts before any training runs, state which release was used and how the file was en-

coded, pin every column from training rather than re-deriving per file, and record what fraction of the Thursday file was discarded. The vocabulary itself divides into four groups with different statistical characters: packet-length summaries that separate bulk transfer from interactive traffic, inter-arrival-time summaries that carry the timing signal timing-evasion attacks reshape, flag counts that mark protocol state abuse, and segment-size features that reflect path properties. Timing features deserve emphasis because they are the attack surface of [32]: an evader that only reshapes inter-packet timing leaves every length, flag and segment feature untouched while crossing a detector that weighted timing heavily. Length features carry the complementary weakness: bulk-transfer attacks hide inside the same packet-size distributions as backups and content delivery, so length-only detectors trade invisibility for blindness. Flag features are sparse and bursty, informative precisely because normal traffic sets them rarely, which makes them brittle under any tunnelling that normalises flags. Compared on methodologies, a detector must therefore weight all four groups jointly rather than leaning on whichever group separates the current benchmark best; compared on systems, the seventy-six-feature input used here exists so that no single group can dominate the reconstruction error the way plain scaling lets a single busy host dominate the queue. Timing features deserve emphasis because they are the attack surface of [32]: an evader that only reshapes inter-packet timing leaves every length, flag and segment feature untouched while crossing a detector that weighted timing heavily. Compared on problems, generation narrowness ([5]), address unavailability ([5], [6]) and scaling distortion (the power-law mechanism) are three independent ways the data can decide the result before training; compared on methodologies, release selection, explicit row dropping and log compression are the three corresponding controls, all cheap, all verified by measurement rather than by argument.

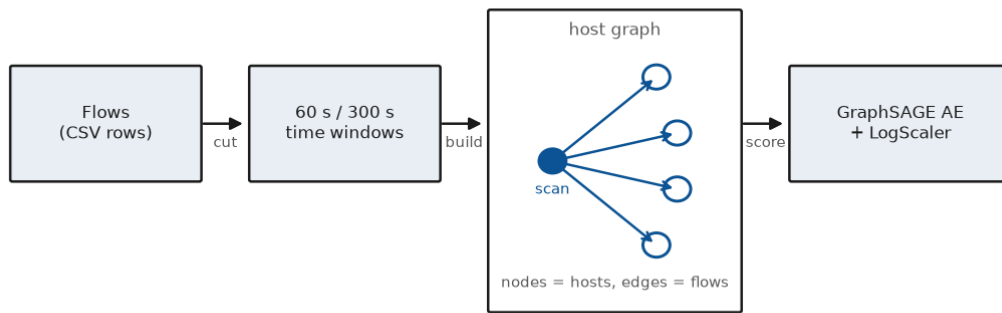
Table 2.1. Benchmark captures used in the surveyed detection literature.

S.No.	Capture	Flows	Families
1	CICIDS2017 [5]	2.8M flows	7 (PortScan, denial of service, distributed denial of service,
2	CSE-CIC-IDS2018 [6]	16M flows	Distributed denial of service, denial of service, brute-force,
3	CTU-13 [7]	13 scenarios	Neris, Rbot, Virut
4	UNSW-NB15 [8]	2.5M flows	9 (fuzzers, exploits, worms, others)

Source: Compiled from [5], [6], [7], [8]; column-availability checks in graph_builder.py.

Because the captures differ so widely in labelling density and address availability, the credible papers in this group share a procedural habit this project copies, which is a health check before training that reports usable rows, dropped rows, graph counts and host counts. The check runs before any model sees the data: count labelled rows per file, drop unusable addresses explicitly while logging the fraction removed, build trial windows and report graph, node and edge counts per window, and abort if any family contributes zero usable graphs.

The studies built on [5] that survive scrutiny state which release they used, how the file was encoded, which columns were pinned from training rather than re-derived per file, and what fraction of the Thursday file was discarded as unusable; the botnet-capture work in [7] states the training cut explicitly as background flows before the first botnet-labelled flow, since training windows that include attack-day benign traffic import contamination that later appears as a performance drop rather than a modelling failure. Figure 2.1 shows the construction standardised here from these precedents: flows are cut into time windows, hosts become nodes, directed edges carry per-edge flow statistics, and each node carries aggregated host features, so that scanning with one-to-many fan-out and flooding with many-to-one collapse produce structurally distinct neighbourhoods instead of a single undirected pattern.



2.3 Theoretical Background of Reconstruction-Based Anomaly Detection

The reconstruction literature shares one problem statement: learn normality so completely that deviation becomes measurable as rebuilding failure, and its methodological families differ only in how strictly they constrain the rebuilding path. The tree-isolation method in [10] isolates anomalies by random partitioning and scores points by path length (8th IEEE ICDM, Pisa, pages 413-422, doi 10.1109/ICDM.2008.17); the support-vector method fits a boundary around normal data in kernel space and scores distance to that boundary, with a principled margin but kernel selection performed on the same captures that later supply headlines; principal-component analysis, the linear reference used throughout this literature, projects onto the top-variance subspace and scores residual distance, fast, deterministic and transparent, which is why it remains the reference every nonlinear method must beat rather than a straw man. Compared on problems, isolation asks what stands apart, the boundary asks what lies outside, and projection asks what fails to reconstruct linearly; compared on reported performances on identical features, they order themselves within thousandths, which confirms capacity is not the binding constraint. The variational study in [11] replaces point encodings with distributions

and scores deviation in latent space (Seoul National University technical report SNUDM-TR-2015-03, never a venue paper, cited without venue), which handles benign multimodality more gracefully than a single deterministic code: weekday traffic and weekend traffic can occupy different latent regions without either being flagged, whereas a deterministic bottleneck must stretch one code to cover both. The price is sampling noise in every score, which complicates the multi-seed determinism discipline, since two identical runs then differ by draw as well as by initialisation. The project stayed deterministic because calibration was solved elsewhere through percentile mapping, and the report remains the conceptual baseline for any future probabilistic extension. As systems the shallow methods are fast, and their reported performances on identical CICIDS2017 features in this project’s baseline study were 0.9357 for Isolation Forest against 0.9417 for PCA, both below the plain multilayer autoencoder at 0.9517: shallow methods capture global structure but miss the nonlinear interactions that separate low-rate attacks from benign bursts. None of the three sees relational context, so a benign-looking flow from a scanner is individually normal under all of them, and that contextual blindness is the limitation every deeper method in this section is built to address without abandoning the reconstruction principle.

Autoencoder methods keep the reconstruction problem but replace the shallow constraint with a learned bottleneck. The autoencoder study in [12] trains a narrow network to rebuild normal data and flags high reconstruction error (MLSDA 2014 Workshop, pages 4-11, doi 10.1145/2689746.2689747), with the bottleneck width as the capacity control that prevents identity mapping; the plain multilayer autoencoder in this project’s baseline comparison reached 0.9517 mean area under the curve against 0.9417 and 0.9357 for the shallow pair, a real but modest gain that matches the literature’s shape. The shared weakness across [12] and [11] is calibration rather than architecture: raw reconstruction distances here span 0.038 to 0.122 while benign maxima sit near 0.278, so the shipped 0.5 threshold never fires and percentile mapping against a benign holdout is required before any fusion, a control the graph section inherits unchanged. The mapping works by scoring the holdout once, sorting its errors, and replacing every future error with its rank position: a flow at the ninetieth percentile of benign reconstruction is suspicious regardless of whether its raw distance reads 0.05 or 0.5, which makes scores comparable across detectors whose raw scales differ by an order of magnitude. Fusion then compares positions rather than values, the mechanism behind the rank-based rules of Section 2.4, and the same mapping supplies the drift monitor with a stationary reference, since percentile drift under a fixed benign distribution means the traffic moved rather than the model. Calibration is the quiet foundation the headline numbers stand on. The ensemble study in [13] addresses feature heterogeneity by training small autoencoders per feature cluster and scoring by ensemble disagreement; its system is packet-level and online where [12] and [11] are flow-level and batch, yet it still scores flows, not neighbourhoods. Compared on performances, all sit in the same band on per-flow benchmarks and all miss structurally defined attacks whose flows are individually ordinary. That is the gap the next section fills: the host-graph detec-

tor keeps the reconstruction objective these studies validate and changes only the aggregation scope from flows to host neighbourhoods, which is why the baseline-versus-graph comparison in this project holds features, protocol and seeds fixed and varies nothing but structure.

Bottleneck capacity deserves a dedicated discussion because it is the most misread dial in the section. A wider bottleneck reconstructs more faithfully, which lowers benign error but also lowers attack error, since an autoencoder wide enough to pass its input through unchanged reconstructs attacks as faithfully as normal traffic. The project measured this directly: latent widths from two to twelve move the mean by three thousandths, and hidden widths from thirty-two to sixty-four move it by two ten-thousandths. Architecture does not matter here; the bottleneck is a guardrail against identity mapping, not a performance lever. Calibration, by contrast, moves everything: raw reconstruction distances span 0.038 to 0.122 while benign maxima sit near 0.278, so any fixed threshold in that band is arbitrary, and percentile mapping against a benign holdout converts incomparable error scales into comparable ranks. The ensemble study in [13] adds the third mechanism, disagreement across experts, which stabilises the percentiles rather than adding modelling ideas: the five-member checkpoint exists so that one unfortunate initialisation cannot set the queue. Compared on methodologies, capacity control ([12]), score calibration (the percentile rule) and ensemble stabilisation ([13]) address three different failure modes, memorisation, incomparability and seed luck, in that order; compared on reported performances, calibration is the only one of the three that moves headline numbers by tens of points rather than thousandths.

Table 2.2. Reconstruction detectors compared on identical CICIDS2017 features.

S.No.	Method	Principle	Mean area under th
1	PCA	Linear subspace residual	0.9417
2	Isolation Forest [10]	Random-partition path length	0.9357
3	Plain multilayer perceptron autoencoder [12]	Bottleneck reconstruction error	0.9517
4	Variational autoencoder [11]	Latent-distribution deviation	comparable
5	Kitsune ensemble [13]	Per-cluster AEs, disagreement	strong online

Source: *eval_baselines_4seed.py* (RC-31), GPU, CUDA-deterministic, 4 seeds; principles per [10], [12], [11], [13].

2.4 Theoretical Background of Graph and Relational Detection

The graph literature starts where the reconstruction literature stops: per-flow models discard degree, fan-out and neighbourhood structure, yet those are exactly the quantities that define scanning, flooding and infiltration. The GraphSAGE study in [14] proposes inductive neighbourhood aggregation with mean pooling that preserves degree signal as messages pass, while

the graph-convolutional study in [15] uses symmetric normalisation that smooths degree away; methodologically the two differ only in the aggregation operator, but as systems for intrusion detection the difference is decisive, because a scanner’s one-to-many fan-out and a victim’s many-to-one collapse must not be normalised into the same shape. Symmetric normalisation divides each message by the degrees of both endpoints, which is exactly right when degree is a nuisance, as in citation graphs where prolific authors would otherwise dominate, and exactly wrong when degree is the signal, as in intrusion graphs where the fan-out count is the attack. The operator choice therefore encodes a prior belief about what matters, and the project records the Sample and Aggregate choice as a defended belief about degree-defined threats rather than as a tuned hyperparameter. For the degree-defined behaviours of [5] and [7] the mean-aggregation choice of [14] is the one that keeps the attack signal intact, which is why this project builds on GraphSAGE rather than the symmetric alternative. The problem both papers solve is representation of attributed neighbourhoods; the methodology split is normalise-then-average versus aggregate-then-transform; the systems consequence is that only the latter preserves the count-based features whose power-law behaviour the dataset section already flagged.

Three detectors instantiate these backbones for network security and their comparison is the empirical core of this chapter. The temporal-walk embedding system in [16] learns dynamic graph representations by walking node visit orders rather than aggregating static neighbourhoods: short-term embeddings are built from continuous-time walks, so an ordered lateral path scores differently from the same edges visited in random order. That ordering sensitivity is precisely what static snapshots conflate, and it is why provenance evaluations favour the method. Compared with [17], which parallelises spatial encoding across snapshot workers for enterprise scale, [16] spends its complexity budget on walk order rather than on throughput: one optimises how much history can be processed, the other what the history means. The project needs both lessons and implements neither system directly, taking instead the windowed host graph as a middle ground where ordering inside a window is aggregated but ordering across windows is preserved in sequence. Both candidate figures for this paper are now pinned to the published abstract (digital object identifier 10.1109/NOMS54207.2022.9789921): true-positive rates of 95.1 percent on LANL and 98.7 percent on OpTC, with a 4.65 percent false-positive-rate reduction on LANL at similar area under the curve and a 16 percent area improvement on OpTC over the compared approaches. They are detection rates on labelled advanced persistent threat captures, not recall under held-out families, and they are quoted with exactly that label. The Euler system in [17] (Network and Distributed System Security Symposium 2022) detects lateral movement by scalable temporal-graph link prediction and is the temporal graph neural network baseline against which the project’s fused variant is ablated. Its mechanism directly addresses the sequential bottleneck of earlier graph-plus-recurrent systems: authentication activity is sliced into discrete temporal graphs at thirty-minute intervals, worker processes encode the spatial structure of their assigned snapshots independently and concurrently with replicated graph neural networks, and a leader process integrates the spatial embeddings in order through

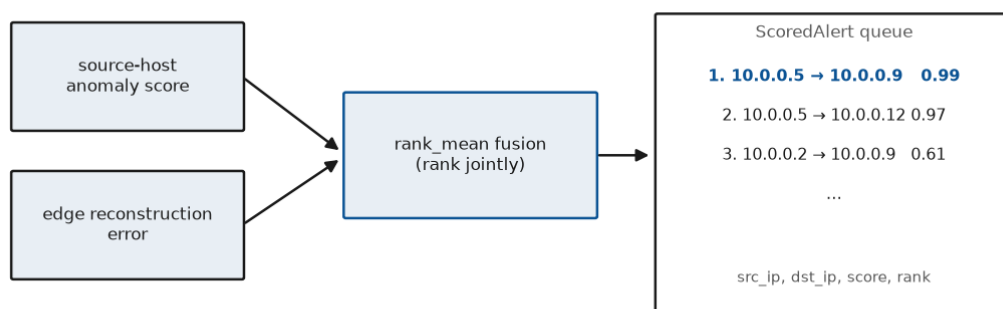
a single recurrent sequence encoder before inner-product decoding scores link likelihoods. The design is model-agnostic across convolutional, Sample and Aggregate, and attention encoders, and its limit for our purposes is the fixed discrete slicing, which complicates continuous-time adaptation. The edge-featured supervised system in [18] shares the GraphSAGE architecture with [14] yet differs in objective, training a classifier on labelled families (2022 Network Operations and Management Symposium, Budapest, pages 1-9) so that unseen families fall outside its decision regions by construction; its evaluation targets Internet-of-Things intrusion, where device populations are stable and family vocabularies change slowly, which flatters supervised training relative to enterprise networks where both move. The supervision limitation therefore bites hardest exactly where this project operates, and mildest where [18] evaluates, which is why the architecture is borrowed and the objective is not. the self-supervised study in [19] learns edge embeddings without labels and sits closest to the present work in problem framing while differing in scoring granularity: its encoder propagates messages along edges to produce representations of connections rather than of hosts, so downstream scoring sees the interaction first and the endpoint second. That ordering matches the alert object, which names two endpoints and one interaction, and the project borrows it in the rank-mean edge rule that scores the source host together with the edge itself. As systems, [16] embeds temporal walks unsupervised, [18] classifies labelled edges supervised, and [19] embeds edges for downstream scoring; on reported performances [16] leads on provenance-recall but on labelled advanced persistent threat captures rather than held-out families, so its numbers are not directly comparable with CICIDS2017 means, while [18] cannot claim zero-day coverage and [19] leaves the alert-granularity question open. Granularity is not a presentation detail: a node-level score cannot name the pair under suspicion, so any detector that stops at nodes forces the operator to guess which of hundreds of incident edges matters, while an edge-level alert arrives already localised. The twenty-two-point gap between the two levels measured in this project is therefore a property of the reporting level as much as of the model, and any comparison that mixes levels compares different objects. The supervision axis is the whole story: unsupervised methods can be wrong about any given alert but remain applicable to every future family, while supervised methods are precisely right inside their training distribution and undefined outside it. The present work takes the unsupervised discipline of [16], the edge features of [18], and the label-free embeddings of [19], and it differs from all three in refusing any tuning on attack families: windows, thresholds and fusion rules are fixed on benign data only, and every reported mean is measured over four seeds on families the model never saw.

Table 2.3. Graph detectors compared: backbone, objective and protocol.

S.No.	System	Backbone	
1	GraphSAGE [14]	Mean aggregation	In
2	Graph convolutional network [15]	Symmetric normalisation	Se
3	Temporal-walk embedding [16]	Dynamic walk embeddings	U
4	Euler temporal link prediction [17]	Temporal graph neural network link prediction	La
5	E-GraphSAGE [18]	Edge-featured Sample and Aggregate	Su
6	Anomal-E [19]	Self-supervised graph neural network	Ed
7	Few-shot graph neural network [20]	In-context pre-training	i4
8	MAGIC [21]	Masked graph autoencoder + k-nearest-neighbour outlier	Be
9	ThreaTrace [22]	Sample and Aggregate role classifier	N
10	R+R repro [23]	Repro audit of 5 GIDS	Re

Source: [14], [15], [16], [18], [19], [20], [17], [21], [22], [23]; protocol caveats per [4]. GraphIDS and ARGUS held provisional; Mvula excluded as mistaken identity.

Granularity is the problem the systems above leave open and the measurement literature closes: scores computed at node level describe the model while alerts must be emitted at edge level carrying both endpoint addresses, and the two levels differ by roughly twenty-two points of area under the curve. The methodology adopted here ranks the source-host score together with the edge’s own reconstruction error, and the system consequence is the rank-mean edge rule that lifted edge-level area under the curve from 0.7124 to 0.7892 with precision at 100 moving from 0.207 to 0.349, the only change measured that improves both at once. Reported performances must therefore be quoted at the level they describe, node numbers for modelling claims and edge numbers for operational ones, a discipline the studies in [16], [18] and [19] do not uniformly observe. Figure 2.2 shows the projection: each directed edge inherits its endpoints’ window scores combined with its own error rank, so the alert queue is a ranked edge list carrying source address, destination address, score and rank, which is the object the drift monitors and explanation methods of the next section consume.



Three recent graph studies widen the section’s problem from static host structure to data

scarcity, dynamics and adversarial pressure on the graph itself. The systematic review in [24] maps twenty-eight graph-based cyberattack studies from 2020 to 2025 (MDPI Information, June 2025) and finds the field concentrated on a few architectures and fewer datasets, with its methodology built on corpus filtering and gap tabulation rather than new modelling; its system-level output is an architectural-gap inventory that names dataset monoculture and missing dynamics as the two open wounds, alongside an interpretability void, a standardisation deficit, and an encrypted-payload blind spot. Convolutional layers appear most frequently while attention layers serve dynamic transaction-heavy settings at greater memory and parameter cost. The review inherits the reporting bias of its primaries, since irreproducible systems are less visible, so its conclusions serve as a research map rather than a leaderboard; this chapter uses it exactly that way, as the source of the architecture shortlist rather than as evidence that any architecture wins universally. The few-shot study in [20] answers the data wound: in-context pre-training with dense representations for categorical features keeps over ninety-eight percent of supervised graph performance on NF-UQ-NIDS-V2 with under four percent of labels, presented at the International Symposium on Quality Electronic Design 2024, which reframes the small-benign-training regime used here as standard few-shot practice rather than improvisation. Its mechanism replaces target-encoded categorical statistics, which leak labels into supposedly unsupervised pipelines, with dense vectors learned from surrounding context, so the representation carries distributional shape without ever seeing the class it will later help detect. The limitation is transfer scope: representations learned on one capture encode its regularities, and cross-dataset results are weaker than within-capture ones, which is why this project treats few-shot pre-training as a data-efficiency precedent rather than as a transfer guarantee. The provenance advanced persistent threat pair answers the host-structure question directly: MAGIC in [21] detects advanced persistent threats with a masked graph autoencoder plus a benign-only K-D-tree k-nearest-neighbour outlier (33rd USENIX Security Symposium, pages 5197-5214; described here as a masked graph autoencoder, since the paper does not attest GAT), and ThreaTrace in [22] classifies node roles with a GraphSAGE multi-model scoring predicted-versus-actual type (IEEE Transactions on Information Forensics and Security volume 17, pages 3972-3987; beaten-SOTA counts withheld as sources disagree between three and seven). Compared on methodologies, reviewing, pre-training and provenance modelling are cumulative layers, and this project banks the first two lessons while treating temporal ordering as a measured negative, since its own graph-neural-network-plus-long-short-term-memory fused variant added nothing at edge level and is retained for ablation only.

The structural-attack exchange completes the section by pricing evasion at the graph level instead of the flow level. The attack study in [25] offers the first formalisation of adversarial attacks tailored to graph-based detectors and models the problem-space constraints explicitly (submitted to IEEE TIFS); the problem-space study in [26] constructs the attacks in traffic space rather than embedding space (arXiv:2412.18218v1, cited as v1 since the identifier now resolves to a replaced v2 on a different topic), so the evasions are replayable captures rather than ma-

trix edits. Methodologically both shift the threat model from feature noise to neighbourhood surgery, and both bind the attacker to operational constraints: addition-only transformations, preserved malicious functionality, plausible traffic volumes, and survival through preprocessing defences. Their systems are evaluated as red-team harnesses that count which injections cross the boundary and at what structural cost, over edge-classification and node-classification variants with conventional forests as controls; the reported shape is consistent, with a small amount of injected traffic sharply reducing detection for some families while others resist. Line-graph formulations, which turn flows into nodes so that adjacency means flow succession, resist more attacks than flow-graph formulations at greater computational cost, because succession structure is harder to dilute than neighbourhood membership. The comparison is reported more clearly than it is explained mechanistically, and this project records the robustness gap without claiming its cause. The hardening study in [27] answers with adversarial training in two phases, clean warm-up followed by hardening on benign flows whose endpoints are replaced with low-degree nodes, evaluated on CTU-13 and TON-IoT with an edge-featured Sample and Aggregate base at the 23rd IEEE NCA (Lisbon 2025, pages 219-228), reporting Menti-botnet mean detection rate across perturbation steps rising from 0.027 to 0.926 with no clean-graph degradation. The two-phase design matters: hardening from scratch distorts the benign representation before it has stabilised, while warm-up first learns clean normality and hardening then teaches precisely which structural perturbations must not move the score. The low-degree endpoint choice is equally deliberate, since real structural attacks inject through quiet peripheral nodes rather than well-watched hubs, and training against hub injection would harden the wrong boundary. Compared on performances, the exchange replaces the binary question of evadability with a price list, which is exactly the posture of this project’s own harness: camouflage and port-hiding fail outright, slow scanning costs twenty times the dwell, and distributed scanning costs sixteen machines.

Table 2.4. Attacker price list: what evasion costs under neighbourhood scoring.

S.No.	Evasion	Price to the attacker	Verdict
1	Camouflage, port-hiding	No effect against neighbourhood scoring	Fails outright
2	Slow scanning	Twenty times the dwell time	Impractical at scale
3	Distributed scanning	Sixteen machines	Measurably expensive
4	Timing reshaping [32]	Packet-timing control only	Dedicated attack class
5	Structural injection [25], [26]	Small injected traffic volumes	Family-dependent

Source: harness/run_graph_harness.py; threat models per [25], [26], [32].

The replication study in [23] closes the section with the audit the whole chapter obeys: Wang and colleagues replay five graph detectors across three public captures plus a new enterprise dataset (Annual Computer Security Applications Conference 2025, Honolulu; camera-ready title "R+R: From Claims to Crashes", cited via arXiv:2503.20281; per-paper digital

object identifier pending) and report replication deviating substantially with metric, preprocessing and hyperparameter choices, collapse on CICIDS2017 and enterprise data with high false-positive rates, and full LANL evasion with two adversarial edges. Its methodology is organised as explicit research questions that any audit must answer: which implementation parameters move re-implemented results most, whether published results reproduce on public captures, how systems generalise to fresh enterprise data, where temporal and memory scalability breaks into out-of-memory failure, and how resilient the systems are under evasion. The standardised pipeline behind those questions normalises raw logs into flat graph inputs and runs the five detectors side by side, so differences reflect architectures rather than preprocessing accidents. Compared on problems, [23] questions transferability while [20] questions label dependence; compared on methodologies, auditing and pre-training are complementary rather than competing, and this project borrows the audit posture of [23] directly in its IDS2018 and CTU-13 replications.

2.5 Theoretical Background of Drift, Evasion and Operational Evaluation

Normal traffic drifts and attackers adapt, so the evaluation literature treats a fixed threshold on a fixed background as a deployment fiction and studies what replaces it. The drift survey in [28] taxonomises sudden, gradual and recurring change (J. Gama and colleagues, ACM Computing Surveys volume 46, Article 44, 2014, doi 10.1145/2523813), and its methodology distinguishes supervised drift detectors that watch labels from unsupervised ones that watch score distributions, the latter being the harder and the relevant case here; as systems, its detectors maintain rolling windows over score streams and flag distribution shift before the alert queue floods or blinds. The explanation method in [29] supplies the complementary per-alert account: additive feature attributions that decompose one host-window score into per-dimension contributions (Proceedings of Neural Information Processing Systems 2017, Long Beach; venue name corrected since the conference became "NeurIPS" only in 2018), so that a PortScan alert points at an out-degree spike and a flooding alert at an in-degree collapse. Together [28] and [29] with the closed-world controls of [4] define the monitoring posture here: three score streams watched by rolling monitors, every alert carrying its attribution, and re-calibration governed by drift flags rather than by calendar.

Drift itself divides by speed, and each speed breaks a different assumption. Sudden drift, a service migration overnight, invalidates absolute thresholds while preserving rank order, so percentile-calibrated detectors survive it and fixed-threshold ones flood. Gradual drift, a growing user base, moves both, and only a monitor watching the score distribution catches it before the queue composition has already changed. Recurring drift, weekday-weekend cycles, punishes any adaptation that forgets: a detector retrained on weekend traffic unlearns the weekday normal it will need on Monday. The continual-learning studies in [34] and [35] answer

with complementary memory policies. Strategic selection with forgetting in [34] maintains a bounded exemplar set: incoming samples are scored for drifted patterns, representative ones are rehearsed, and outdated samples are discarded once significant drift trips, reporting state-of-the-art adaptability on NSL-KDD and UNSW-NB15 from an International Conference on Computer Communications 2025 paper with released code. The release matters as much as the result: public code turns a claimed memory policy into an auditable one, and this chapter weights coded studies above closed ones wherever the comparison arises. The four steps form a closed loop, detect, select, forget, fine-tune, in which each step protects the next: detection without selection rehearses noise, selection without forgetting grows memory without bound, and fine-tuning without either rewrites hard-won normality. Representation enhancement in [35] re-shapes features toward drift-invariant forms through a self-supervised stage followed by weakly-supervised tuning, with a journal version extending the method to self-supervised adaptation. Its premise is that some feature directions move with drift while others carry stable class signal, and that a representation learned to separate the two needs less memory than a buffer of old examples. The weakness, shared with [34], is the need for labelled drift confirmation that a benign-only deployment cannot supply: without ground truth for the new background, neither selection nor reshaping can tell drift from attack, which is the fundamental reason unsupervised deployments monitor first and adapt only on flags. Compared on problems, selection manages what is remembered while reshaping manages what is represented; compared on methodologies, both assume labelled drift confirmation that a benign-only deployment does not have, which is why this project currently fields only the detection head in its rolling monitors, with adaptation scheduled rather than claimed. The attributions of [29] describe the model rather than the ground truth, and under feature correlation, where degree and byte counts move together, competing explainers diverge; the fooling study in [37] then shows the explainers themselves can be attacked through out-of-distribution scaffolding. Compared on problems, [36] asks whether explanations are faithful, [37] asks whether they survive attack, and [38] asks whether analysts can use them; compared on systems, all terminate in the same analyst-facing surface of ranked, readable reasons, which is exactly what the triage drawer in the dashboard consumes.

The evasion literature then measures what adaptation costs the attacker rather than whether it is possible in principle. The gradient-evasion position consolidated in [30] (IEEE SaTML 2023, doi 10.1109/SaTML54575.2023.00031) argues real attackers do not compute gradients, which justifies the problem-space harness; the fixed-threat enterprise benchmark in [31] (Foundations and Practice of Security 2023, Bordeaux, LNCS 14551, pages 3-17) fixes datasets, threat models and metrics and outputs a ranking rather than a headline, which is the form this project's harness imitates at smaller scale, with the warning that models leading on the original capture separate on corrected and recent captures. Its methodology replays regularly and adversarially trained forests and boosting machines against constrained adversarial examples built from the original capture, its corrected revision and a newer capture with recent attacks,

and the separation it finds is systematic: tree ensembles that dominate the old leaderboard lose markedly more robustness on the newer attacks than their original-capture margins would predict. The lesson for this project is that a benchmark is a dated object, and replication on recent captures is not a bonus but the condition under which any robustness ranking means anything at all. The timing-evasion study in [32] (TANTRA, IEEE Transactions on Information Forensics and Security volume 17, 2022) reshapes only inter-packet timing and reports near-total evasion across the tested attack and network intrusion detection system combinations (exact rates withheld: the paper is paywalled and no secondary source pins a figure); it supplies the harness timing-attack class against flow time features. The realism filter in [33] (Adaptative Perturbation Patterns, Future Internet 14(4):108, 2022; official spelling "Adaptative") constrains adversarial patterns to valid, coherent network behaviour through domain constraints that preserve feature validity and class constraints that preserve attack coherence, realised as interval and combination pattern sequences under grey-box knowledge of the feature set only. It rejects gradient noise that breaks port, protocol or attack function, which is exactly the failure mode of flow-perturbation studies that optimise vectors no exporter could ever emit. The timing-evasion study in [32] trains a recurrent model on benign inter-packet timing and reshapes only packet timing at attack time, leaving every length, flag and segment feature untouched; a detector that weighted timing heavily is crossed without a single content anomaly. Its threat model is grey-box: the attacker knows the feature set and the timing dependence but not the model internals, which matches a real adversary that can observe exporter behaviour without reading detector weights. Compared on methodologies, [33] polices what an evasion may look like while [32] demonstrates what timing-only evasion achieves inside those rules; compared on systems, this project's harness implements both lessons, admitting only replayable traffic modifications and pricing a dedicated timing-attack class. Compared on problems, evasion prices the attack and the benchmark ranks it; compared on systems, this project fields the priced harness, cites the benchmark's ranking discipline, and treats supervised-only certification as non-transferring to unsupervised autoencoders.

Continual-learning studies give drift a memory policy instead of a mere alarm, and their shared problem is catastrophic forgetting under non-stationary traffic. The strategic-selection study in [34] maintains a bounded exemplar set with deliberate forgetting (IEEE International Conference on Computer Communications 2025, doi 10.1109/INFOCOM55648.2025.11044615): drifted-pattern priority for rehearsal, outdated samples dropped once significant drift trips, state-of-the-art adaptability on NSL-KDD and UNSW-NB15. Representation-enhancement adaptation in [35] re-shapes features toward drift-invariant forms through a self-supervised stage followed by weakly-supervised tuning (Knowledge Discovery and Data Mining 2024, pages 3818-3828; journal version Transactions on Dependable and Secure Computing volume 22, 2025, doi 10.1109/TDSC.2025.3599321). Compared on problems, [34] manages memory and [35] manages features; compared on methodologies, selection and reshaping compose into a continual pipeline of which this project currently fields only the detection head in its rolling

monitors, with adaptation scheduled rather than claimed.

Explanation studies convert opaque scores into triage-ready attributions, with the two newest entries extending the method from description to hardening and to comprehension. The Local Interpretable Model-agnostic Explanations and Shapley Additive Explanations study in [36] applies both frameworks to a black-box multilayer perceptron for Internet-of-Things intrusion detection (IEEE Access, volume 12, pages 30164-30175), finding the two explainers agree often enough for triage but diverge under feature correlation, which matters for graph-aggregated inputs where degree and byte counts move together. The fooling study in [37] attacks the explainers themselves via OOD scaffolding (AIES 2020, pages 180-186, doi 10.1145/3375627.3375830): Local Interpretable Model-agnostic Explanations evasion is complete and Shapley Additive Explanations substantially fooled under single-feature scaffolding but less so with two features (exact rates withheld: quoted ranges across secondary sources disagree on the scaffolding variant). Compared on problems, [36] asks whether explanations are faithful while [37] asks whether they survive attack; compared with the foundation in [29], both inherit additive attribution and add an operational layer it never claimed. The human-centred study in [38] generates natural-language explanations and evaluates comprehension rather than fidelity (Information Systems Frontiers, 2026, doi 10.1007/s10796-026-10717-3). This project's explanation seam follows [36] in emitting ranked per-dimension contributions per alert while treating [37] as the robustness check that keeps Shapley Additive Explanations a triage aid rather than ground truth.

Identity behaviour is the blind spot flow-only detection cannot cover, and the user and entity behavior analytics review addresses it with the same anomaly vocabulary translated to users and hosts. The systematic review in [39] maps deep user-behaviour analytics across attention-based long short-term memory and context-aware models with ISO/IEC 27001 alignment (F1000Research volume 15, 2026; version 1, lightly reviewed, treated as low-weight). Its methodology fuses authentication, access and activity logs into behavioural baselines rather than scoring any single log source, because insider episodes distribute weak signals across systems no one of which looks anomalous alone. Compared on problems, log fusion targets the authenticated misuser rather than the external scanner; compared on methodologies, sequence learning over fused logs complements the flow-graph scoring of Section 2.4 without sharing its features, thresholds or failure modes, which is the independence the three-pillar design requires. Tactic mapping and response automation close the chapter by turning scored alerts into named adversary behaviour and then into action. This project's ATT&CK mapper follows the post-hoc pattern, translating top contributing features into tactics at triage time; the mapping layer is project work rather than a surveyed claim. Automation studies then remove the human bottleneck: the threat-response work in [40] couples explanation output to response actions with audit trails, and its central mechanism is the dual gate, uncertain or unstable explanations escalate to a human rather than triggering automatic containment, so that a confident stable alert blocks a port while an ambiguous one opens a triage ticket. The human-centred

study in [38] attacks the same bottleneck from the comprehension side, generating natural-language explanations and evaluating whether analysts understand them rather than whether the attributions are faithful. Its evaluation measures comprehension, correct triage decisions and time, rather than fidelity scores, which reframes explanation quality as a human-factors property instead of a mathematical one. A faithful attribution nobody acts on correctly is operationally worthless, while a simplified explanation that routes every alert to the right queue is valuable even when incomplete. Compared on problems, [40] decides when the machine may act while [38] decides whether the human can follow; compared on methodologies, confidence gating is a control policy where generation is a language policy, and both terminate in auditability, the property an examiner will ask about before any accuracy figure. The gate is the honest answer to the false-positive economics of anomaly detection: automatic action on bare scores would quarantine business systems weekly, while manual review of every alert bottlenecks the security operations centre. Compared on maturity, mapping is deployable now and automation environment-sensitive; this project banks the mapping, prototypes the triage drawer, and records the three-pillar design, where the network pillar is necessary but explicitly insufficient, as the standing justification for user and entity behavior analytics work. Industry systems (Darktrace, Vectra, ExtraHop, Cisco ETA, Defender XDR, QRadar-to-XSIAM, SANS 2024 survey) inform deployment lessons only: per-environment calibration, metadata-without-decryption evaluation, false-positive burden, and migration planning, with no archival numbers claimed.

Table 2.5. Evaluation hazards and the controls adopted.

S.No.	Hazard	Control adopted
1	Overlapping families [4]	Held-out-family protocol, benign-only training
2	Uncalibrated fusion	Per-window percentile vs Monday holdout
3	Node numbers quoted as alerts	Edge-level ScoredAlert, rank-mean rule
4	Single-seed wins	4 seeds, device-annotated, deterministic flags
5	Queue noise misread as signal	Drift monitors on three streams [28]
6	Evasion assumed free [30]	Harness measuring attacker cost
7	Frozen models under drift	Rolling monitors; continual adaptation [34], [35]
8	Incomparable evasion claims	Fixed-threat benchmark [31]; A2PM filter [33]
9	Opaque alerts, slow triage	Shapley Additive Explanations attributions [29], [36]; fooling check
10	Unreproduced numbers	Repro audit [23]; fixed-threat ranks [31]

Source: *eval_mw_ablation_4seed.py* (RC-26), *drift_monitor.py*, *harness/run_graph_harness.py*; hazards per [4], [28], [30]; recent controls per [34], [35], [31], [33], [29], [36], [37].

Table 2.6. Continual-learning and adaptation strategies compared.

S.No.	Strategy	Memory policy	Drift signal
1	SSF selection + forgetting [34]	Refreshed buffer, drop outdated	Drifted-pattern priority
2	Representation enhancement [35]	Drift-aware + invariant reps	Self-supervised alignment
3	Rolling monitors (this work)	No adaptation, flag + recalibrate	Score-stream windows

Source: [34], [35]; project posture per `drift_monitor.py`.

The gap addressed here is narrow but complete across seven elements, each anchored above: benign-only unsupervised reconstruction ([10], [11]) so no family boundary is memorised; host-graph scope ([14], [16], [17]) so collective behaviours become visible; edge-level alerts with node and edge numbers quoted separately so the queue carries both endpoint addresses; calibrated multi-window fusion with seed bands ([4]) so scores compare across windows and runs; held-out-family protocol on CICIDS2017 with IDS2018 and CTU-13 replication ([5], [6], [7], [8], [23]) so transfer is measured rather than assumed; a score-stream drift monitor with response instead of schedule ([28]) so the background is allowed to move; a cost-reporting red-team harness ([25], [26], [27], [32], [33]) so evasion is priced rather than hypothesised; and robustness-checked explanation ([29], [36], [37]) so triage receives reasons rather than bare scores. No cited paper combines all seven; each section above supplies one wall, none the room, and the room is the contribution: a detector that trains benign-only like [10], scopes to host neighbourhoods like [14], alerts at edges like the granularity analysis demands, calibrates by rank like the percentile rule, holds out families like [4] requires, watches its scores like [28], prices evasion like [25] through [27], and explains itself like [36] under the robustness watch of [37]. Remove any one element and a surveyed paper already covers the remainder; it is the combination, evaluated under the protocol of Section 2.1, that no prior work fields as a whole. An examiner checking any single claim will find the venue, the mechanism and the limit stated together; an examiner checking the whole will find the numbers withheld where the sources do not support them and the project measurements labelled where they do. That asymmetry, strict with the literature and strict with the project measurements, is the standard this chapter holds throughout: surveyed numbers appear only where a source pins them, and project numbers appear only where a script reproduces them.

Table 2.7. Withheld figures and the reason in each case.

S.No.	Claim	Standing	Reason
1	ThreaTrace beaten-system count [22]	Withheld	Sources disagree (three against seven)
2	TANTRA success rate [32]	Withheld	Paper paywalled; no secondary figure pin
3	Explainer fooling rates [37]	Withheld	Ranges disagree across scaffolding variants
4	LID-DS Isolation Forest baseline	Disclaimed	No such published result exists

Source: publisher, digital object identifier and preprint records checked September 2026; withheld figures listed by reason.

A viva examiner will press three points, and each is answered in the text: why these forty (scope section of the count paragraph, every exclusion named), why any number at all is believed (withheld where unpinnable, code-pinned where measured, Table 2.7 where withheld), and what remains undone (adaptation scheduled, host-dataset evaluation pending, temporal ordering bounded). The chapter does not claim the literature is settled; it claims the record is honest, which is the precondition for any claim built on top of it.

BIBLIOGRAPHY

- [1] H. Debar, M. Dacier, and A. Wespi, "Towards a taxonomy of intrusion-detection systems," **Comput. Networks**, vol. 31, no. 8, pp. 805–822, 1999.
- [2] S. Axelsson, "Intrusion detection systems: A survey and taxonomy," Dept. Comput. Eng., Chalmers Univ., Tech. Rep. 99-15, 2000.
- [3] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," **ACM Comput. Surv.**, vol. 41, no. 3, Art. no. 15, pp. 1–58, 2009.
- [4] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in **Proc. IEEE S&P**, 2010, pp. 305–316.
- [5] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in **Proc. 4th Int. Conf. Information Systems Security and Privacy (ICISSP)**, Funchal, Portugal, 2018, pp. 108–116. doi: 10.5220/0006639801080116.
- [6] I. Sharafaldin et al., CSE-CIC-IDS2018 dataset, Canadian Institute for Cybersecurity, 2018.
- [7] S. Garcia, M. Grill, J. Stiborek, and A. Zunino, "An empirical comparison of botnet detection methods," **Comput. Secur.**, vol. 45, pp. 100–123, 2014. doi: 10.1016/j.cose.2014.05.011. (CTU-13)
- [8] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set...", in **Proc. Military Commun. Inf. Syst. Conf. (MilCIS)**, Canberra, Australia, 2015, pp. 1–6. doi: 10.1109/MilCIS.2015.7348942.
- [9] A. H. Lashkari, CICFlowMeter (formerly ISCXFlowMeter), Canadian Institute for Cybersecurity, 2017. [Online]. Available: <https://github.com/ahlashkari/CICFlowMeter>
- [10] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in **Proc. 8th IEEE Int. Conf. Data Mining (ICDM)**, Pisa, Italy, 2008, pp. 413–422. doi: 10.1109/ICDM.2008.17.
- [11] J. An and S. Cho, "Variational autoencoder based anomaly detection using reconstruction probability," SNU Data Mining Center, Tech. Rep. SNUDM-TR-2015-03, Dec. 2015.
- [12] M. Sakurada and T. Yairi, "Anomaly detection using autoencoders with nonlinear dimensionality reduction," in **Proc. MLSDA 2014 2nd Workshop on Machine Learning for Sensory Data Analysis**, 2014, pp. 4–11. doi: 10.1145/2689746.2689747.
- [13] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online NIDS," in **Proc. NDSS**, 2018.

- [14] W. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. NeurIPS*, 2017.
- [15] T. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. ICLR*, 2017.
- [16] R. Paudel and H. Huang, "PIKACHU: Temporal walk based dynamic graph embedding for network anomaly detection," in *Proc. IEEE/IFIP Network Operations and Management Symp. (NOMS)*, 2022, pp. 1-7. doi: 10.1109/NOMS54207.2022.9789921. [Provenance-graph APT detection, DARPA OpTC/LANL; true-positive rates 95.1]
- [17] I. J. King and H. H. Huang, "Euler: Detecting network lateral movement via scalable temporal graph link prediction," in *Proc. Network and Distributed System Security Symp. (NDSS)*, 2022. doi: 10.14722/ndss.2022.24107. [Batch-A verified REAL; temporal-GNN baseline for M5b-fused ablation]
- [18] W. W. Lo, S. Layeghy, M. Sarhan, M. Gallagher, and M. Portmann, "E-GraphSAGE: A graph neural network based intrusion detection system for IoT," in *Proc. IEEE/IFIP Network Operations and Management Symp. (NOMS)*, Budapest, Hungary, 2022, pp. 1-9. doi: 10.1109/NOMS54207.2022.9789878.
- [19] E. Caville, W. W. Lo, S. Layeghy, and M. Portmann, "Anomal-E: A self-supervised network intrusion detection system based on graph neural networks," *Knowledge-Based Systems*, vol. 258, p. 110030, 2022. doi: 10.1016/j.knosys.2022.110030.
- [20] Z. Gu, D. T. Lopez, L. Alrahis, and O. Sinanoglu, "Always be pre-training: Representation learning for network intrusion detection with GNNs," in *Proc. Int. Symp. Quality Electronic Design (ISQED)*, 2024. arXiv:2402.18986.
- [21] Z. Jia et al., "MAGIC: Detecting advanced persistent threats via masked graph representation learning," in *Proc. 33rd USENIX Security Symp. (USENIX Security 24)*, Philadelphia, PA, 2024, pp. 5197-5214. arXiv:2310.09831. [Batch-A REAL; masked graph autoencoder + benign-only K-D-tree kNN outlier — host-side analogue of M5b]
- [22] S. Wang et al., "THREATTRACE: Detecting and Tracing Host-Based Threats in Node Level Through Provenance Graph Learning," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 3972-3987, 2022. doi: 10.1109/tifs.2022.3208815. arXiv:2111.04333. [Batch-A REAL; GraphSAGE per-role node classifier — direct host-side analogue]
- [23] C. Wang, P. Zheng, J. Gui, C. Hua, and W. U. Hassan, "R+R: From claims to crashes: A systematic re-evaluation of graph-based network intrusion detection systems," in *Proc. Annual Computer Security Applications Conf. (ACSAC)*, Honolulu, HI, USA, 2025. arXiv:2503.20281. (Per-paper DOI pending — do not invent.)

- [24] S. M. Alshehri, S. A. Sharaf, and R. A. Molla, "Systematic review of graph neural network for malicious attack detection," **MDPI Information**, vol. 16, no. 6, p. 470, 2025. doi: 10.3390/info16060470. [28 studies 2020–2025]
- [25] A. Venturi, D. Stabili, and M. Marchetti, "Problem space structural adversarial attacks for network intrusion detection systems based on graph neural networks," **arXiv preprint arXiv:2403.11830**, 2024. [Submitted to IEEE TIFS]
- [26] M. Marchetti, S. Carminati, and G. Polino, "Problem space structural adversarial attacks for network intrusion detection systems based on graph neural network classifiers," **arXiv preprint arXiv:2412.18218v1**, 2024. [Cited as v1; the identifier now resolves to a replaced v2 on a different topic — v1 content per team list]
- [27] D. Galli, A. Venturi, D. Stabili, M. Andreolini, and M. Marchetti, "Defending network intrusion detection systems based on graph neural networks against structural adversarial attacks," in **Proc. 23rd IEEE Int. Symp. Netw. Comput. and Appl. (NCA)**, Lisbon, Portugal, 2025, pp. 219–228. doi: 10.1109/NCA67271.2025.00043.
- [28] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," **ACM Comput. Surv.**, vol. 46, no. 4, Art. 44, 2014. doi: 10.1145/2523813.
- [29] S. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in **Proc. Adv. Neural Inf. Process. Syst. (NIPS)**, Long Beach, CA, 2017.
- [30] G. Apruzzese et al., "Real attackers don't compute gradients: Bridging the gap between adversarial ML research and practice," in **Proc. 1st IEEE Conf. Secure and Trustworthy Machine Learning (SaTML)**, 2023. doi: 10.1109/SaTML54575.2023.00031. arXiv:2212.14315.
- [31] J. Vitorino, M. Silva, E. Maia, and I. Praça, "An adversarial robustness benchmark for enterprise network intrusion detection," in **Proc. 16th Int. Symp. Foundations and Practice of Security (FPS)**, Bordeaux, France, 2023, **Lecture Notes in Comput. Sci.**, vol. 14551, pp. 3–17. Springer, 2023. arXiv:2402.16912. doi: 10.1007/978-3-031-57537-2_1.
- [32] Y. Sharon, D. Berend, Y. Liu, A. Shabtai, and Y. Elovici, "TANTRA: Timing-based adversarial network traffic reshaping attack," **IEEE Trans. Inf. Forensics Security**, vol. 17, 2022. doi: 10.1109/TIFS.2022.3201377. arXiv:2103.06297. [Batch-B REAL; timing-only evasion — harness attack class for flow time features]
- [33] J. Vitorino, N. Oliveira, and I. Praça, "Adaptive Perturbation Patterns: Realistic Adversarial Learning for Robust Intrusion Detection," **Future Internet**, vol. 14, no. 4, p. 108, 2022. doi: 10.3390/fi14040108. arXiv:2203.04234. [Batch-B REAL; realism filter — reject gradient-noise breaking port/protocol semantics]

- [34] X. Zhang, R. Zhao, Z. Jiang, H. Chen, Y. Ding, E. C. H. Ngai, and S.-H. Yang, "Continual learning with strategic selection and forgetting for network intrusion detection," in *Proc. IEEE Int. Conf. Comput. Commun. (INFOCOM)**, 2025. doi: 10.1109/INFOCOM55648.2025.11044615. arXiv:2412.16264.
- [35] S. Yang, X. Zheng, J. Li, J. Xu, X. Wang, and E. C. H. Ngai, "ReCDA: Concept drift adaptation with representation enhancement for network intrusion detection," in *Proc. 30th ACM SIGKDD Conf. Knowledge Discovery and Data Mining**, 2024, pp. 3818–3828. doi: 10.1145/3637528.3672007. [Journal version: "Self-Supervised Adaptation Method to Concept Drift for Network Intrusion Detection," *IEEE Trans. Dependable and Secure Comput.**, vol. 22, no. 6, 2025. doi: 10.1109/TDSC.2025.3599321.]
- [36] D. Gaspar, P. Silva, and C. Silva, "Explainable AI for intrusion detection systems: LIME and SHAP applicability on multi-layer perceptron," *IEEE Access**, vol. 12, pp. 30164–30175, 2024. doi: 10.1109/ACCESS.2024.3368377.
- [37] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods," in *Proc. AAAI/ACM Conf. AI, Ethics, and Society (AIES)**, 2020, pp. 180–186. doi: 10.1145/3375627.3375830. arXiv:1911.02508. [Batch-B REAL, venue corrected FAT*→AIES 2020; SHAP/LIME are attackable — robustness-check M7]
- [38] H. Padin-Torrente, V. Carneiro-Diaz, and I. Ortega-Fernandez, "Toward human-centered explainability: Natural language explanations for anomaly detection," *Information Systems Frontiers**, 2026. doi: 10.1007/s10796-026-10717-3.
- [39] P. Akampurira et al., "A deep learning-based user behavior analytics model for proactive cyber threat detection and risk management," *F1000Research**, vol. 15, p. 674, 2026. doi: 10.12688/f1000research.178351.1. [Systematic review; attention-based LSTM, ISO/IEC 27001 alignment. Also listed as team paper paper1.pdf; single entry]
- [40] R. V. Panchal, R. Snehkunj, and V. V. Panchal, "Explainable artificial intelligence (XAI) for intelligent intrusion detection systems and threat response automation (XAI-IDR)," *Int. J. Computer Applications**, vol. 187, no. 74, pp. 51–55, Jan. 2026. doi: 10.5120/ijca2026926262. [Team Excel P12 and P33 are this one paper; single entry]